

Product.java

```
package EcommercePlatformSearch;

public class Product {

    //productId, productName, and category.

    private int productId;

    private String productName;

    private String category;

    public Product(int productId, String productName, String category) {

        this.productId = productId;

        this.productName = productName;

        this.category = category;

    }

    public String getCategory() {

        return category;

    }

    public String getProductName() {

        return productName;

    }

}
```

```
public int getProductId() {  
    return productId;  
}  
  
@Override  
public String toString() {  
    return "Product ID: " + productId +  
        ", Name: " + productName +  
        ", Category: " + category;  
}  
  
}
```

LinearSearch.java

```
package EcommercePlatformSearch;  
  
public class LinearSearch {  
  
    public static Product linearSearch(Product[] products, int targetId) {  
        for (Product product : products) {  
            if (product.getProductId() == targetId) {  
                return product;  
            }  
        }  
    }  
}
```

```
return null;
```

```
}
```

BinarySearch.java

```
package EcommercePlatformSearch;
```

```
public class BinarySearch {  
    public static Product binarySearch(Product[] products, int targetId) {  
  
        int left = 0;  
  
        int right = products.length - 1;  
  
        while (left <= right) {  
  
            int mid = left + (right - left) / 2;  
  
            if (products[mid].getProductId() == targetId) {  
                return products[mid];  
            } else if (products[mid].getProductId() < targetId) {  
                left = mid + 1;  
            } else {  
                right = mid - 1;  
            }  
        }  
        return null;  
    }  
}
```

Main.java

```
package EcommercePlatformSearch;

import java.util.Scanner;

import static EcommercePlatformSearch.BinarySearch.binarySearch;
import static EcommercePlatformSearch.LinearSearch.linearSearch;

public class Main {

    static void main(String[] args) {

        Product[] products = new Product[]{
            new Product(1 , "Laptop", "Electronics"),
            new Product(2 , "Mobile", "Electronics"),
            new Product(3 , "Refrigerator", "Electronics"),
            new Product(4 , "Shoe", "Fashion"),
            new Product(5, "Watch", "Accessories")
        };

        Scanner sc = new Scanner(System.in);

        int targetId = sc.nextInt();

        sc.nextLine();

        Product result1 = linearSearch(products, targetId);
```

```
System.out.println("Linear Search Result:");
```

```
System.out.println(result1);
```

```
// Binary Search
```

```
Product result2 = binarySearch(products, targetId);
```

```
System.out.println("\nBinary Search Result:");
```

```
System.out.println(result2);
```

```
}
```

```
}
```

```
|
```